



7 企业综合项目实战

7.1 数据准备

1. 将项目提供的4个文件，放到 `/home/hewwen8888/data` （注意放到自己权限的目录下）目录下

```
ch7_course_info.txt  ch7_stu_info.txt  
ch7_score_info.txt   ch7_teacher_info.txt
```

2. 建表语句：

```
-- 创建学生表
DROP TABLE IF EXISTS ds_hive.ch7_stu_info;
create table if not exists ds_hive.ch7_stu_info(
    stu_id string COMMENT '学生id',
    stu_name string COMMENT '学生姓名',
    sex string COMMENT '性别'
)
row format delimited fields terminated by ','
stored as textfile
;

-- 创建课程表
DROP TABLE IF EXISTS ds_hive.ch7_course_info;
create table if not exists ds_hive.ch7_course_info(
    course_id string COMMENT '课程id',
    course_name string COMMENT '课程名',
    tea_id string COMMENT '任课老师id'
)
row format delimited fields terminated by ','
stored as textfile;

-- 创建老师表
DROP TABLE IF EXISTS ds_hive.ch7_teacher_info;
create table if not exists ds_hive.ch7_teacher_info(
    tea_id string COMMENT '老师id',
    tea_name string COMMENT '老师姓名'
)
row format delimited fields terminated by ','
stored as textfile
;

-- 创建分数表
DROP TABLE IF EXISTS ds_hive.ch7_score_info;
create table if not exists ds_hive.ch7_score_info(
    stu_id string COMMENT '学生id',
    course_id string COMMENT '课程id',
    score int COMMENT '成绩'
)
row format delimited fields terminated by ','
stored as textfile
```

```
;
```

3. 插入数据

```
load data local inpath '/home/hewwen8888/data/ch7_stu_info.txt' overwrite into table ds_hive.ch7_s
load data local inpath '/home/hewwen8888/data/ch7_course_info.txt' overwrite into table ds_hive.ch
load data local inpath '/home/hewwen8888/data/ch7_teacher_info.txt' overwrite into table ds_hive.c
load data local inpath '/home/hewwen8888/data/ch7_score_info.txt' overwrite into table ds_hive.ch7
```

7.2 简单查询

1. 检索课程名称为“hadoop”且分数小于60的学生的课程信息，结果按分数降序排列

```
select t2.*
       ,t1.score
from ds_hive.ch7_score_info t1

join ds_hive.ch7_course_info t2
on t1.course_id=t2.course_id
where t1.score<60 and t2.course_name='hadoop'
order by t1.score desc
;
```

运行结果：

	t2.course_id	t2.course_name	t2.tea_id	t1.score
1	02	hadoop	1001	58
2	02	hadoop	1001	55
3	02	hadoop	1001	52
4	02	hadoop	1001	48
5	02	hadoop	1001	44
6	02	hadoop	1001	39
7	02	hadoop	1001	39
8	02	hadoop	1001	35
9	02	hadoop	1001	34
10	02	hadoop	1001	34

2. 查询hive成绩不及格的学生姓名和其对应的成绩，按照学生id升序排序

```
select t1.stu_id
      ,t3.stu_name
      ,t1.score
from ds_hive.ch7_score_info t1
join ds_hive.ch7_course_info t2
  on t1.course_id=t2.course_id
join ds_hive.ch7_stu_info t3
  on t1.stu_id=t3.stu_id
where t1.score<60 and t2.course_name='hive'
order by t1.stu_id
;
```

运行结果：

	t1.stu_id	t3.stu_name	t1.score
1	006	钟小雨	59
2	008	范军	39
3	010	赵梦	23
4	014	李建国	32
5	017	陈晓	55
6	018	王当然	49

3. 查询学生“小明”的课程的成绩

```
SELECT t1.stu_id
,t2.stu_name
,sum(t1.score) as sum_score
from ds_hive.ch7_score_info t1
join ds_hive.ch7_stu_info t2
on t1.stu_id=t2.stu_id
where t2.stu_name='小明'
group by t1.stu_id,t2.stu_name
;
```

运行结果：

查询历史记录
保存的查询
查询生成器
结果 (1)

	t1.stu_id	t3.stu_name	score
1	001	小明	290

4. 查询各科成绩最高和最低的分，以如下的形式显示：课程号，最高分，最低分

```
SELECT t1.course_id
,max(t1.score) as max_score
,min(t1.score) as min_score
from ds_hive.ch7_score_info t1
group by t1.course_id
;
```

运行结果:

历史记录		保存的查询	查询生成器	结果 (5)
course_id		max_score	min_score	
1	03	99	23	
2	01	98	38	
3	04	100	34	
4	02	93	34	
5	05	83	59	

5. 查询男生、女生人数

```
SELECT t1.sex
,count(*) as sex_cnt
from ds_hive.ch7_stu_info t1
group by t1.sex
;
```

运行结果:

		sex	stu_cnt
1		男	11
2		女	9

6. 查询平均成绩大于60分的学生的姓名和平均成绩

```
SELECT t2.stu_name
,avg(t1.score) as avg_score
from ds_hive.ch7_score_info t1
left join ds_hive.ch7_stu_info t2
on t1.stu_id=t2.stu_id
group by t2.stu_name
having avg_score>=60
;
```

运行结果：

	t1.stu_id	t3.stu_name	avg_score
	004	小刘	81.5
	006	钟小雨	71.33333333333333
	010	赵梦	68.75
	016	范当然	81.25
	019	邓紫	65.66666666666667
	001	小明	72.5
	002	小李	86.5
	005	刘鹏飞	74.8
	011	周磊	81
0	015	陈丽华	70.25
1	020	张晓	68
2	009	赵静	73.6
3	013	张晓明	61

7. [课堂讲解]查询每门课程的平均成绩，结果按平均成绩升序排序，平均成绩相同时，按课程号降序排列

```
SELECT t1.course_id
,avg(score) as avg_s
FROM ds_hive.ch7_score_info t1
group by t1.course_id
sort by avg_s asc,t1.course_id desc
;
```

运行结果：

	course_id	avg_score
1	02	60.63157894736842
2	03	68.15789473684211
3	04	68.91666666666667
4	01	69.84210526315789
5	05	72.6

8. 查询学生的总成绩并按照总成绩降序排序

```
SELECT t1.stu_id
, sum(t1.score) as sum_score
from ds_hive.ch7_score_info t1
group by t1.stu_id
order by sum_score desc
;
```

运行结果：

	t1.stu_id	t3.stu_name	sum_score
1	005	刘鹏飞	374
2	009	赵静	368
3	002	小李	346
4	004	小刘	326
5	016	范当然	325
6	007	赵梦婷	299
7	001	小明	290
8	015	陈丽华	281
9	010	赵梦	275
10	020	张晓	272
11	013	张晓明	244
12	011	周磊	243
13	018	王当然	228
14	006	钟小雨	214

9. 按照如下格式显示学生的hive、spark、hadoop三科成绩和平均成绩，没有成绩的输出为0，按照学生平均成绩降序显示

Stu_name		Avg_s	Hive_s	Spark_s	Hadoop_s
001	0	89	78	...	

分析：行转列（后续会有专门函数讲解），平均，排序

```

select t1.stu_id
      ,t3.stu_name
      ,avg(t1.score) as avg_score
      ,sum(case when t2.course_name='hive' then t1.score end) as f_hive
      ,sum(case when t2.course_name='spark' then t1.score end) as f_spark
      ,sum(case when t2.course_name='hadoop' then t1.score end) as f_hadoop
from ds_hive.ch7_score_info t1
join ds_hive.ch7_course_info t2
  on t1.course_id=t2.course_id
join ds_hive.ch7_stu_info t3
  on t1.stu_id=t3.stu_id
group by t1.stu_id
      ,t3.stu_name
order by avg_score desc
;

```

```

-- with as 教学喜欢用
with avg_score_info as(
select t1.stu_id
      ,t3.stu_name
      ,t1.score
      ,t2.course_name
from ds_hive.ch7_score_info t1
join ds_hive.ch7_course_info t2
  on t1.course_id=t2.course_id
join ds_hive.ch7_stu_info t3
  on t1.stu_id=t3.stu_id
)
select stu_id,stu_name
      ,avg(score) as avg_score
      ,sum(case when course_name='hive' then score end) as f_hive
      ,sum(case when course_name='spark' then score end) as f_spark
      ,sum(case when course_name='hadoop' then score end) as f_hadoop
from avg_score_info
group by stu_id,stu_name

```

注释： with as 就类似于一个视图或临时表，可以用来存储一部分的sql语句作为别名，不同的是 with as 属于一次性的，而且必须要和其他sql一起使用才可以！其最大的好处就是适当的提

高代码可读性，而且如果with子句在后面要多次使用到，这可以大大的简化SQL；更重要的是：一次分析，多次使用，这也是为什么会提供性能的地方，达到了“少读”的目标（未进行表缓存）。

运行结果：

	t1.stu_id	t3.stu_name	avg_score	f_hive	f_spark	f_hadoop
1	002	小李	86.5	21.75	25	21
2	004	小刘	81.5	22.25	14.75	23.25
3	016	范当然	81.25	17.75	23.5	22.25
4	011	周磊	81	23.333333333333332	0	25
5	005	刘鹏飞	74.8	19.8	17	8.8
6	009	赵静	73.6	12	15.8	15.6
7	001	小明	72.5	19.75	13.5	15.75
8	006	钟小雨	71.33333333333333	19.666666666666668	0	30
9	015	陈丽华	70.25	21	0	12
10	010	赵梦	68.75	5.75	25	17
11	020	张晓	68	20.25	12.5	13

7.3 复杂查询

11. 查询一共参加三门课程考试，且其中一门为hive课程的学生的id和姓名

```

select t1.stu_id
      ,t3.stu_name
      ,count(course_id) as course_cnt
from ds_hive.ch7_score_info t1
join ds_hive.ch7_stu_info t3
  on t1.stu_id=t3.stu_id
where t1.stu_id in (select distinct t1.stu_id
                    from ds_hive.ch7_score_info t1
                    join ds_hive.ch7_course_info t3
                      on t1.course_id=t3.course_id
                    where t3.course_name='hive')

group by t1.stu_id,t3.stu_name
having course_cnt=3
;

```

```

-- 或者
with timu11_ch7_stu_info as (
select distinct t1.stu_id
  from ds_hive.ch7_score_info t1
join ds_hive.ch7_course_info t3
  on t1.course_id=t3.course_id
where t3.course_name='hive'

)

select t1.stu_id
      ,t3.stu_name
      ,count(course_id) as course_cnt
from ds_hive.ch7_score_info t1
join ds_hive.ch7_stu_info t3
  on t1.stu_id=t3.stu_id
where t1.stu_id in (select stu_id from timu11_ch7_stu_info)
group by t1.stu_id
having course_cnt=3
;

```

运行结果：

查询历史记录

保存的查询

查询生成器

结果 (5)

t1.stu_id		t3.stu_name	course_cnt
1	006	钟小雨	3
2	019	邓紫	3
3	011	周磊	3
4	012	王芳	3
5	008	范军	3

12. 查询所有课程成绩均小于60分的学生的学号、姓名

```

SELECT
    t1.stu_id
    ,T3.stu_name
    ,COUNT(CASE WHEN score<=60 THEN course_id END) AS course_CNT --这是一个条件表达式，它会逐个检查成
from ds_hive.ch7_score_info t1
join ds_hive.ch7_stu_info t3
on t1.stu_id=t3.stu_id
GROUP BY t1.stu_id,T3.stu_name
HAVING course_CNT=0
;

```

运行结果：

查询历史记录

保存的查询

查询生成器

结果 (2)

	t1.stu_id	t3.stu_name	course_cnt
1	008	范军	0
2	017	陈晓	0

13. [课堂讲解]查询没有学全所有课的学生的学号、姓名

```

SELECT
T4.stu_id
,T4.stu_name
FROM
(SELECT
    t1.stu_id
    ,T3.stu_name
    ,COUNT(course_id ) AS course_CNT
from ds_hive.ch7_score_info t1
    join ds_hive.ch7_stu_info t3
        on t1.stu_id=t3.stu_id
GROUP BY t1.stu_id,T3.stu_name) T4
JOIN(SELECT COUNT(*) course_CNT FROM ds_hive.ch7_course_info) T5
ON 1=1
WHERE T4.course_CNT<>T5.course_CNT -- <> 是"不等于"运算符。
;

```

```

-- 因为有的学生没学，如果想保留用此sql
SELECT
T4.stu_id
,T4.stu_name
,t4.course_cnt
FROM
(SELECT
    t3.stu_id
    ,T3.stu_name
    ,coalesce(count(t1.course_id),0) as course_cnt
from ds_hive.ch7_stu_info t3
left join ds_hive.ch7_score_info t1
    on t1.stu_id=t3.stu_id
GROUP BY t3.stu_id,T3.stu_name) T4
JOIN(SELECT COUNT(*) course_CNT FROM ds_hive.ch7_course_info) T5
ON 1=1
WHERE T4.course_CNT<>T5.course_CNT -- <> 是"不等于"运算符。
;

```

运行结果：

	t4.stu_id	t4.stu_name
1	004	小刘
2	006	钟小雨
3	010	赵梦
4	014	李建国
5	016	范当然
6	017	陈晓
7	019	邓紫
8	008	范军
9	013	张晓明
10	018	王当然
11	001	小明
12	002	小李
13	011	周磊
14	012	王芳

14. 查询有两门以上的课程不及格的同学的学号及其平均成绩

```
select stu_id
      ,count(case when score<60 then course_id end) as course_cnt
      ,avg(score) avg_score
from ds_hive.ch7_score_info
group by stu_id
having course_cnt>=2
;
```

运行结果：

	stu_id	course_cnt	avg_score
1	007	2	59.8
2	008	3	43
3	013	2	61
4	014	3	48
5	015	2	70.25
6	017	4	45.25
7	018	3	57
8	020	2	68

15. 查询出每门课程的及格人数和不及格人数，按课程名称展示

```
select t3.course_name
,t2.pass_cnt
,t2.nopass_cnt
from
(
select t1.course_id
,count(case when t1.score>=60 then stu_id end) as pass_cnt
,count(case when t1.score<60 then stu_id end) as nopass_cnt
from ds_hive.ch7_score_info t1
group by t1.course_id) t2
join ds_hive.ch7_course_info t3
on t2.course_id=t3.course_id
;
```

运行结果：

t1.course_id	t3.course_name	notpass_cnt	pass_cnt
01	java	6	13
02	hadoop	10	9
03	hive	6	13
04	spark	5	7
05	flink	1	4

16. 查询所有课程成绩在70分以上的学生的姓名、课程名称和分数，按分数升序排列

```

with timu15_score_info_tmp as(
select t2.stu_id
,t2.course_id
,t2.score
from ds_hive.ch7_score_info t2
join(select t1.stu_id
,sum(case when t1.score>=70 then 0 else 1 end) as course_cnt
from ds_hive.ch7_score_info t1
group by t1.stu_id
having course_cnt=0) t3
on t2.stu_id=t3.stu_id
)

select t2.stu_name
,t3.course_name
,t1.score
from timu15_score_info_tmp t1
join ds_hive.ch7_stu_info t2
on t1.stu_id=t2.stu_id
join ds_hive.ch7_course_info t3
on t1.course_id=t3.course_id
order by t1.score
;

```

运行结果：

	t2.stu_name	t4.course_name	t1.score
1	范当然	hive	71
2	范当然	java	71
3	小李	java	75
4	小李	hadoop	84
5	小李	hive	87
6	范当然	hadoop	89
7	范当然	spark	94
8	小李	spark	100

17. 查询学过编号为“01”的课程并且也学过编号为“02”的课程的学生的学号、姓名

```
select t2.stu_id
,t3.stu_name
from
(
select t1.stu_id
,count(course_id) as course_cnt
from ds_hive.ch7_score_info t1
where course_id in ('01','02')
group by t1.stu_id
having course_cnt=2) t2
join ds_hive.ch7_stu_info t3
on t2.stu_id=t3.stu_id
;
```

运行结果：

t2.stu_id		t3.stu_name
1	001	小明
2	002	小李
3	004	小刘
4	005	刘鹏飞
5	006	钟小雨
6	007	赵梦婷
7	008	范军
8	009	赵静
9	010	赵梦
10	011	周磊
11	012	王芳
12	013	张晓明
13	014	李建国
14	015	陈丽华
15	016	范当然
16	017	陈晓
17	018	王当然
18	019	邓紫
19	020	张晓

18. 查询学过“1002”老师所教的所有课的同学的学号、姓名

```

select
t2.stu_id
,t2.stu_name
from
(select
t1.stu_id
,t3.stu_name,count(t1.course_id) course_cnt
from ds_hive.ch7_score_info t1
join ds_hive.ch7_stu_info t3
on t1.stu_id=t3.stu_id
where t1.course_id in (select course_id from ds_hive.ch7_course_info where tea_id='1002')
group by t1.stu_id
,t3.stu_name
) t2
join (select count(course_id) as course_cnt from ds_hive.ch7_course_info where tea_id='1002') t4
on t2.course_cnt=t4.course_cnt
;

```

运行结果:

查询历史记录

保存的查询

查询生成器

结果 (3)

	t2.stu_id	t2.stu_name
1	005	刘鹏飞
2	007	赵梦婷
3	009	赵静

19. 查询至少有一门课与学号为“001”的学生所学课程相同的学生的学号和姓名

```
select t1.stu_id
,t2.stu_name
,count(t1.course_id) as c_cnt
from ds_hive.ch7_score_info t1
left join ds_hive.ch7_stu_info t2
on t1.stu_id=t2.stu_id
where t1.course_id in (select distinct course_id from ds_hive.ch7_score_info where stu_id='001')
and t2.stu_id <> '001'
group by t1.stu_id
,t2.stu_name
;
```

运行结果：

[查询历史记录](#)[保存的查询](#)[查询生成器](#)[结果 \(65\)](#)

	t1.stu_id	t3.stu_name
1	002	小李
2	004	小刘
3	005	刘鹏飞
4	006	钟小雨
5	007	赵梦婷
6	008	范军
7	009	赵静
8	010	赵梦
9	011	周磊
10	012	王芳
11	013	张晓明
12	014	李建国
13	015	陈丽华
14	016	范当然

20. 按平均成绩从高到低显示所有学生的所有课程的成绩以及平均成绩

```
with timu2_avg_score_tmp as (  
select t1.stu_id  
,avg(score) as avg_score  
from ds_hive.ch7_score_info t1  
group by t1.stu_id  
)  
select t1.stu_id  
,t2.stu_name  
,t3.course_name  
,t4.avg_score  
from ds_hive.ch7_score_info t1  
join ds_hive.ch7_stu_info t2  
on t1.stu_id=t2.stu_id  
join ds_hive.ch7_course_info t3  
on t1.course_id=t3.course_id  
join timu2_avg_score_tmp t4  
on t1.stu_id=t4.stu_id  
;
```

运行结果：

	t2.stu_id	t2.score	t4.avg_score
1	002	100	86.5
2	002	87	86.5
3	002	84	86.5
4	002	75	86.5
5	004	93	81.5
6	004	89	81.5
7	004	85	81.5
8	004	59	81.5
9	016	94	81.25
10	016	89	81.25
11	016	71	81.25
12	016	71	81.25
13	011	98	81
14	011	75	81